Capstone 6 – Presentation (Slide Content) and Final Demo Script

Use this in Gamma (free plan): paste the slide text into "Paste in text", or build the slides by hand. The demo script follows after the slides.

Slide 1 – Title

AI Coding Tutor: Can AI Explain Beginner Python Correctly? Capstone 6 – your name, course, date

Slide 2 – The problem

- Beginners now learn code from AI explanations.
- Explanations sound confident even when wrong.
- Question: where does AI mislead learners?

Slide 3 – Method

- 10 beginner programs: loops, functions, lists, dictionaries, files, strings
- AI prompts: explain, trace, practice problems
- Every claim run in Google Colab
- Tools: ChatGPT, Gemini, Claude, Colab, GitHub, Gamma

Slide 4 – Findings

- 10 of 10 programs ran correctly
- 15 claims tested: 10 correct, 5 wrong
- Mistakes cluster in: range/slice ends, list shifts, file modes, shared default arguments
- Scanner flags Python 2 syntax, `eval`, `os.system`, files without `with`

Slide 5 – Automation

- Colab cell `check_answer()` tests a learner's function against hidden cases
- Example: off-by-one `sum_to` scored 1/4 and showed exactly which cases failed
- `verify_claim()` compares AI output claims with real output

Slide 6 – Before vs after

- Before: read code alone. After: AI + run + checker + mini-guide
- Same 10-question quiz both times
- Result: average ___ → ___ (insert your data)

Slide 7 – Responsible AI and conclusion

- Treat AI as a hypothesis, then verify
- Disclose what AI wrote
- Never run unsafe suggestions (`eval`, `os.system`)
- Conclusion: AI is a good first explainer, not a final authority

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Final Demo Script (about 5 minutes)

Time	Show	Say
0:00	Slide 1–2	State the question
0:30	Colab notebook, Cell 1	"This runs any program and captures real output." Run P09
1:15	Chat tool with prompt PR3 on P09	Show the AI's output claim
1:45	Cell 2 `verify_claim()`	Paste the claim; show CORRECT or WRONG
2:30	Cell 3 `scan_risks()`	Paste `x = eval(input())`; show the warning
3:15	Cell 4 `check_answer()`	Run a wrong `sum_to`, then fix it and show 4/4
4:00	Quiz cell 5	Run `grade_quiz()` with a sample answer list
4:30	Slide 6–7	Give before/after result and conclusion

Backup plan: keep screenshots of every cell's output in case Colab is slow.

Requirements checklist

Requirement	Where met
10 short programs	Document 1

Requirement	Where met
Explain each line, trace output, practice problems	Document 1, prompts PR1–PR6
Run in Colab and confirm claims	Documents 2 and 7
Wrong explanations, outdated syntax, unsafe code recorded	Document 3
Mini-guide and 10-question quiz	Documents 4 and 5
Automate one step	Document 11 (cells 4, 5)
Before vs after comparison	Document 8
6+ AI skills	AI foundations, prompt engineering, content writing, research and verification, automation, responsible AI
4+ AI tools	ChatGPT, Gemini, Claude, Google Colab (plus GitHub, Gamma) – list only tools you really used